



SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005

AUTOLIV ASP, INC.
Autoliv North America
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CALIBRATION

Valid To: April 30, 2017

Certificate Number: 0403.16

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations¹:

I. Dimensional Testing/Calibration¹

Parameter/Equipment	Range	CMC ² (±)	Comments
Diameter ³ – Inside/Outside	Up to 12 in	310 µin	Quickvision
Diameter ³ – Inside	(0.22 to 20.36) mm; (0.000866 to 0.802) in	930 µin	Pin gages
Length ³ – X-axis Y-axis Z-axis	Up to 24 in Up to 38 in Up to 22 in	510 µin 710 µin 480 µin	CMM
Length ³ – 1D	Up to 18 in	450 µin	Calipers
Length ³ – 1D	Up to 2 in	70 µin	Micrometers

II. Mechanical

Parameter/Equipment	Range	CMC ^{2,5} (±)	Comments
Load Cell (Non-Seatbelt)	(444 to 44 482) lb	0.21 % FS	SWI Instr. 006 – section 1
Load Cell (Seatbelt)	(2244 to 22 241) lb	1.5 % FS	SWI Instr. 006 – section 8
Displacement Transducer	(1 to 1000) mm	0.70 % FS	SWI Instr. 006 – section 2
Pressure Sensor and Gauge	(5 to 10 000) psi	0.60 % FS	SWI Instr. 006 – section 3
Accelerometer	20 Hz @ 5g; (>30 to 4000) Hz @ 10 g	4.8 % FS	SWI Instr. 006 – section 4

III. Thermodynamics

Parameter/Equipment	Range	CMC ² (±)	Comments
Temperature	(20 & 40) °C	0.66 °C	SWI Instr. 006 – section 5
Relative Humidity	(33 & 75) % RH	1.5 % RH	SWI Instr. 006 – section 5

IV. VSR Equipment⁴

Parameter/Equipment	Range	CMC ^{2,5} (±)	Comments
Dimensional –			
Angle – Measure	(0 to 90)°	0.34°	SWI calibration 001, 003, 004
Bore – Measure	Up to 1.5 mm	0.12 mm	SWI calibration 007
Distance/Displacement – Measure	Up to 2000 mm	2.2 mm	SWI calibration 001 – 004, 006, 008, 009
Electrical – DC/Low Frequency			
AC/DC Voltage – Measure	(0 to 24) V _{DC}	0.15 % FS	SWI calibration 005
Mechanical –			
Acceleration	(0.68 to 70) g/s	12 % FS	SWI calibration 001, 002, 003
Force – Measure	(0 to 8) lbf	0.37 % FS	SWI calibration 002, 005
Mass – Generate	(0 to 70) lb	1.2 % FS	SWI calibration 001 – 003, 004, 006, 009
Pressure – Generate	(0 to 90) psi	3.4 psi	SWI calibration 001 - 003, 005, 007, 008

¹ This laboratory offers commercial calibration service and dimensional testing.

² Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs uncertainties represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC uncertainty due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

³ This laboratory meets R205 – *Specific Requirements: Calibration Laboratory Accreditation Program* for the types of dimensional tests listed above and is considered equivalent to that of a calibration.

⁴ The noted parameters are not offered for commercial service, only offered to Autoliv locations.

⁵ In the statement of CMC, FS is defined as Full Scale.



Accredited Laboratory

A2LA has accredited

AUTOLIV ASP, INC.

Auburn Hills, MI

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets any additional program requirements in the field of calibration. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009).



Presented this 8th day of June 2015.

A handwritten signature in black ink, reading 'Peter Mlynar'.

President & CEO
For the Accreditation Council
Certificate Number 0403.16
Valid to April 30, 2017

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.